HSE 103 Final Project - Gene Expression

Research question

How does gene expression for *CXCR2*, *CYP24A1*, and *PTX3* vary with maternal vitamin D concentration and pregnancy outcome?

The continuous covariate is maternal vitamin D (ng/mL). The two categorical variables are pregnancy outcome and vitamin D group (<30 versus >=30 ng/mL).

Import and prepare the original data

```
# Read the three course CSV files without editing them.
# These files need to be in the same folder as this R Markdown file.
library(tidyverse)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
covariates <- read.csv("GSE85307_covar-1.csv")
expression <- read.csv("GSE85307_geneExpression.csv", row.names = 1)
linker <- read.csv("GSE85307_geneNames.csv")

# The expression file has genes in rows and samples in columns.
# Transpose it so sample IDs become rows and probe IDs become columns.
expression <- t(expression)

# Inspect sample IDs before combining the datasets.
# In the RStudio Console, compare these IDs if you need to check ordering:
# head(covariates$geoID)
# head(rownames(expression))

# Clinical columns contain a description before the colon.
# Separate the description from each recorded value.
covariates <- separate(covariates, mat_vitD,
                       into = c(NA, "vitamin_d"), sep = ": ")
covariates <- separate(covariates, preg_preeclampsia,
                       into = c(NA, "outcome"), sep = ": ")

# Set the types used in the plots.
covariates$vitamin_d <- as.numeric(covariates$vitamin_d)
covariates$outcome <- factor(covariates$outcome,
                             levels = c("Normal", "Preeclampsia"))
covariates$vitamin_d_group <- factor(
```

```
  ifelse(covariates$vitamin_d < 30, "Below 30", "30 or higher"),
  levels = c("Below 30", "30 or higher"))
# Clean maternal age and BMI
covariates <- separate(covariates, mat_age_yrs,
                       into = c(NA, "age_years"), sep = ": ")

covariates <- separate(covariates, mat_BMI,
                       into = c(NA, "bmi"), sep = ": ")

covariates$age_years <- as.numeric(covariates$age_years)
covariates$bmi <- as.numeric(covariates$bmi)

# Start one data frame with one row per clinical sample.
analysis <- data.frame(
  geoID = covariates$geoID,
  vitamin_d = covariates$vitamin_d,
  outcome = covariates$outcome,
  vitamin_d_group = covariates$vitamin_d_group,
  age_years = covariates$age_years,
  bmi = covariates$bmi
)

# Find each gene's probe ID in the linker file.
# Select expression values using the sample IDs, not original row positions.
gene_names <- c("CXCR2", "CYP24A1", "PTX3")
for (gene in gene_names) {
  probe <- as.character(linker$PROBEID[linker$GENE == gene])
  analysis[, gene] <- expression[covariates$geoID, probe]
}

# Inspect the analysis data frame before plotting.
# In the RStudio Console, inspect the prepared data before plotting:
# head(analysis)
```

Function to generate three plots for any selected gene

The function takes two inputs: a data frame and a vector of one or more gene names. It uses a `for` loop to make a histogram, scatterplot, and grouped boxplot for each gene.

```
make_gene_plots <- function(data, gene_names) {
  # Repeat the same three plot types for each selected gene.
  for (gene in gene_names) {

    # Put the selected gene expression and covariates into one plot data frame.
    plotData <- data.frame(
      Expression = data[, gene],
      VitaminD = data$vitamin_d,
      Outcome = data$outcome,
      VitaminDGroup = data$vitamin_d_group
    )

    # Histogram: distribution of expression for this gene.
```

```
    histogram <- ggplot(plotData, aes(x = Expression)) +
      geom_histogram(binwidth = 0.10, fill = "steelblue", color = "white") +
     labs(
  title = gene,
  subtitle = "Expression distribution",
  x = "Gene expression (log2 signal)",
  y = "Number of participants"
) +
theme_classic() +
theme(plot.title = element_text(face = "italic"))

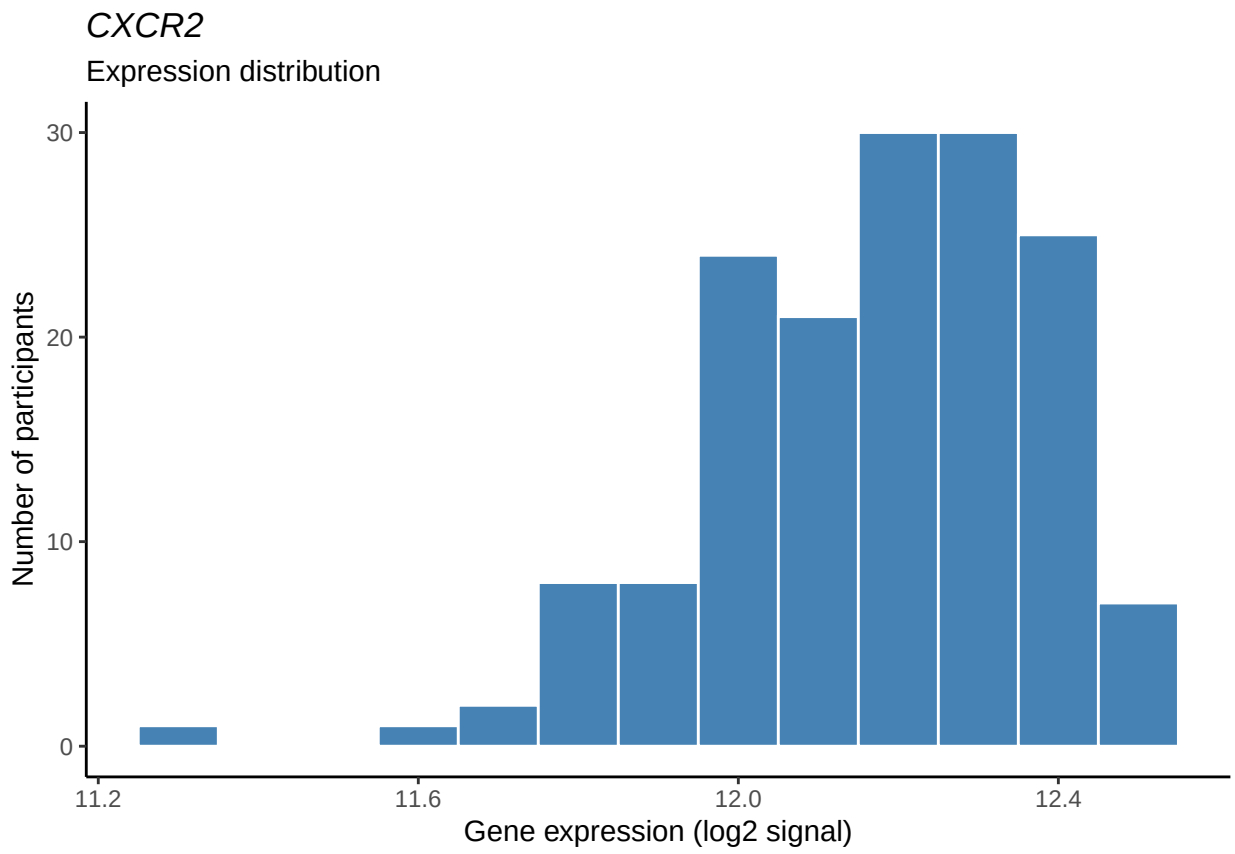
    # Scatterplot: vitamin D on the x-axis and expression on the y-axis.
    scatterplot <- ggplot(plotData, aes(x = VitaminD, y = Expression)) +
      geom_point(color = "steelblue", alpha = 0.7) +
      geom_smooth(method = loess) +
     labs(
  title = gene,
  subtitle = "Expression and maternal vitamin D",
  x = "Maternal vitamin D (ng/mL)",
  y = "Gene expression (log2 signal)"
) +
theme_classic() +
theme(plot.title = element_text(face = "italic"))

    # Boxplot: pregnancy outcome separated by the two vitamin D groups.
    boxplot <- ggplot(plotData, aes(x = Outcome, y = Expression,
                                    fill = VitaminDGroup)) +
      geom_boxplot(position = position_dodge(0.9)) +
      scale_fill_manual(values = c("steelblue", "orange")) +
     labs(
  title = gene,
  subtitle = "Expression by pregnancy outcome",
  x = "Pregnancy outcome",
  y = "Gene expression (log2 signal)",
  fill = "Vitamin D group"
) +
theme_classic() +
theme(plot.title = element_text(face = "italic"))

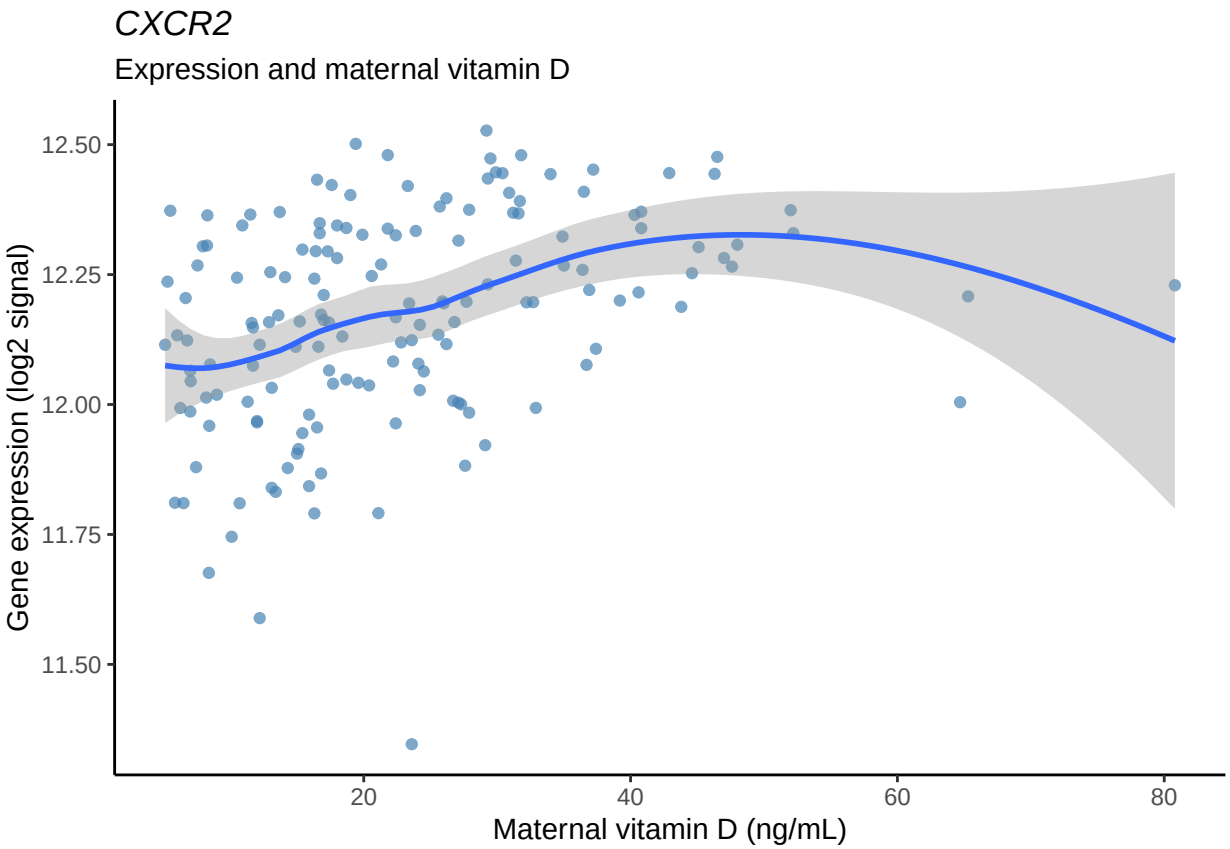
    # print() displays plots made inside a for loop in the knitted report.
    print(histogram)
    print(scatterplot)
    print(boxplot)
  }
}
```

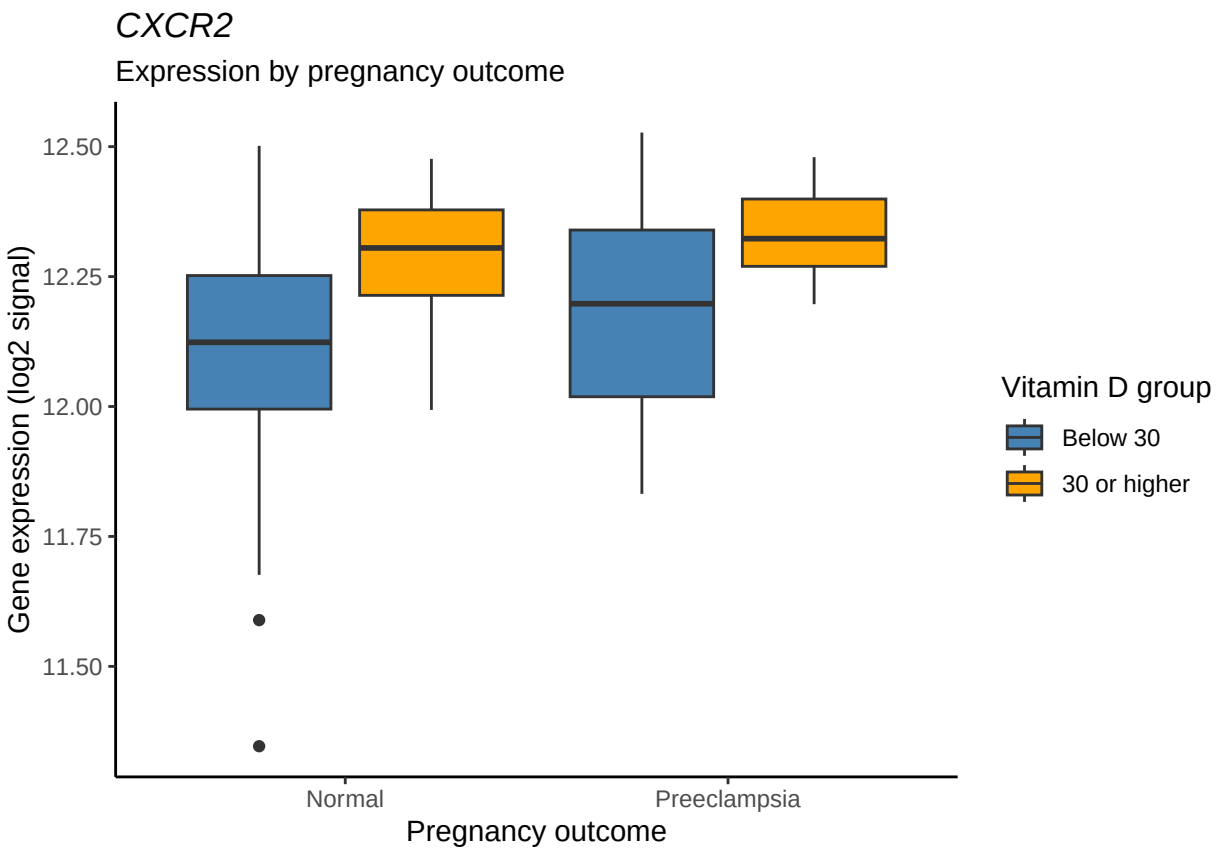
Nine plots for three genes

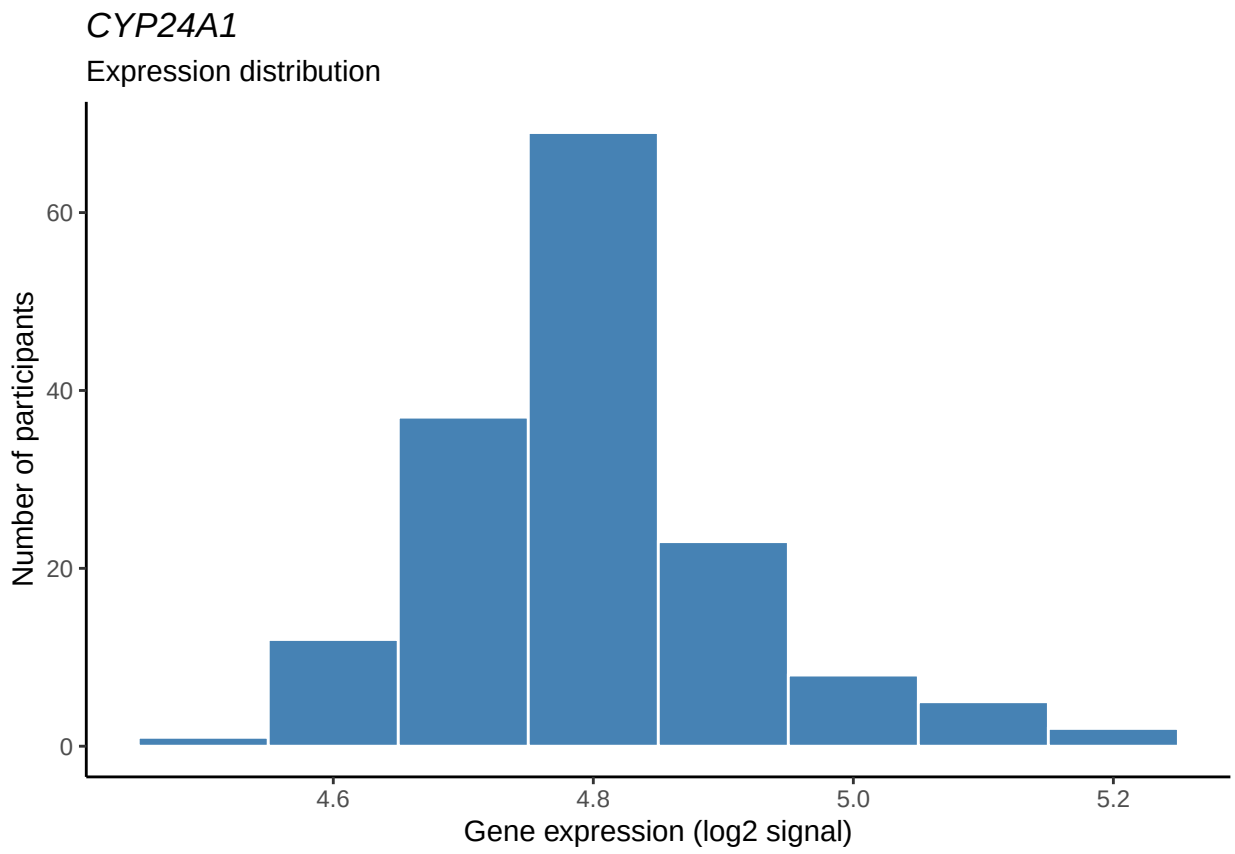
```
# Three genes, each with one histogram, one scatterplot, and one boxplot.
make_gene_plots(analysis, gene_names)
```



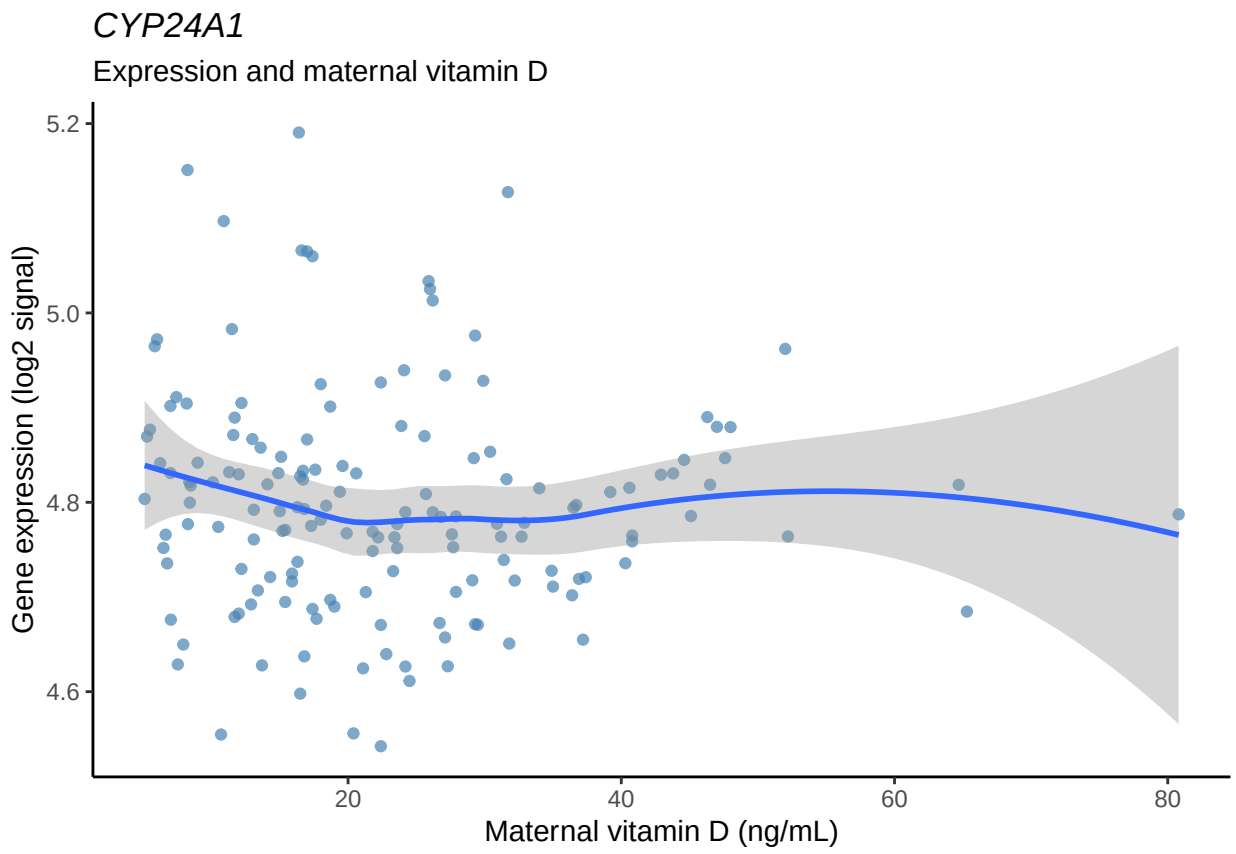

```
## `geom_smooth()` using formula = 'y ~ x'
```

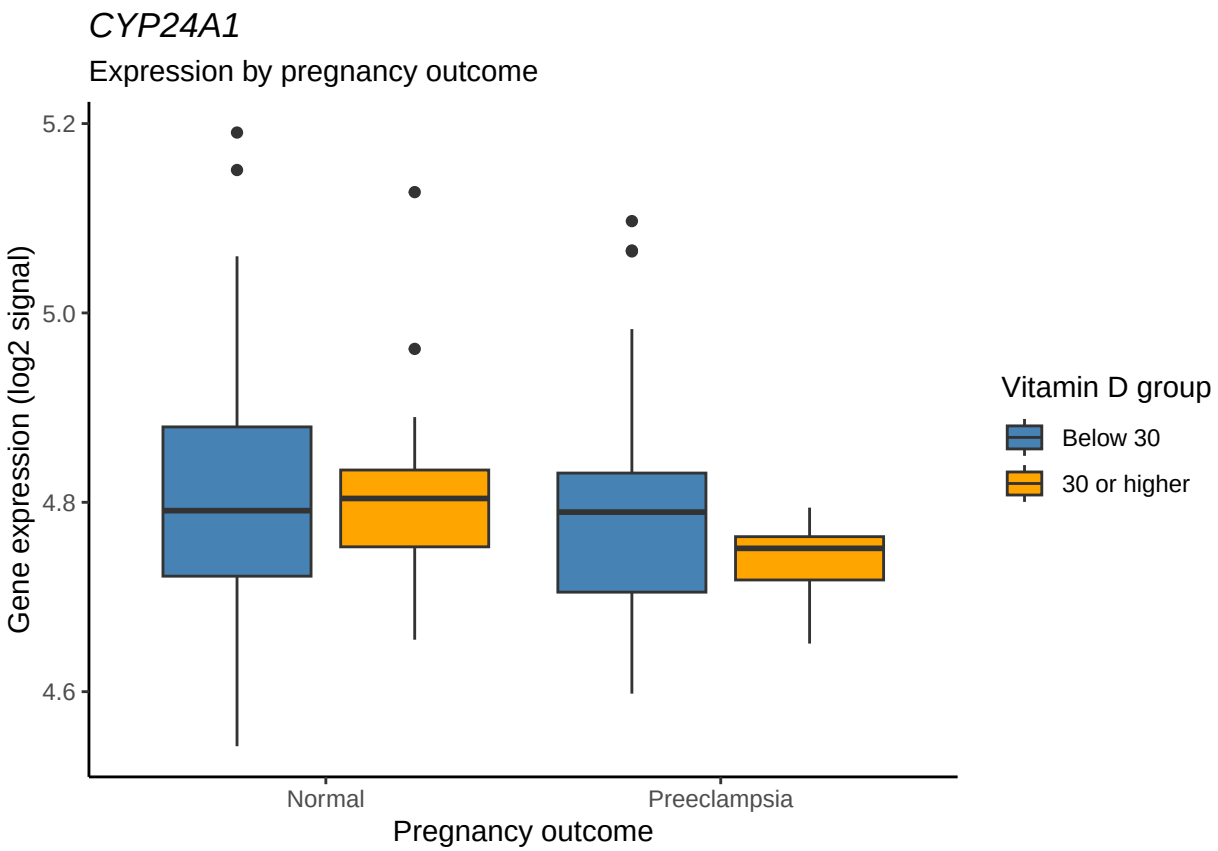

CXCR2
Expression and maternal vitamin D
Gene expression (log2 signal)
12.50
12.25
12.00
11.75
11.50
20
40
60
80
Maternal vitamin D (ng/mL)

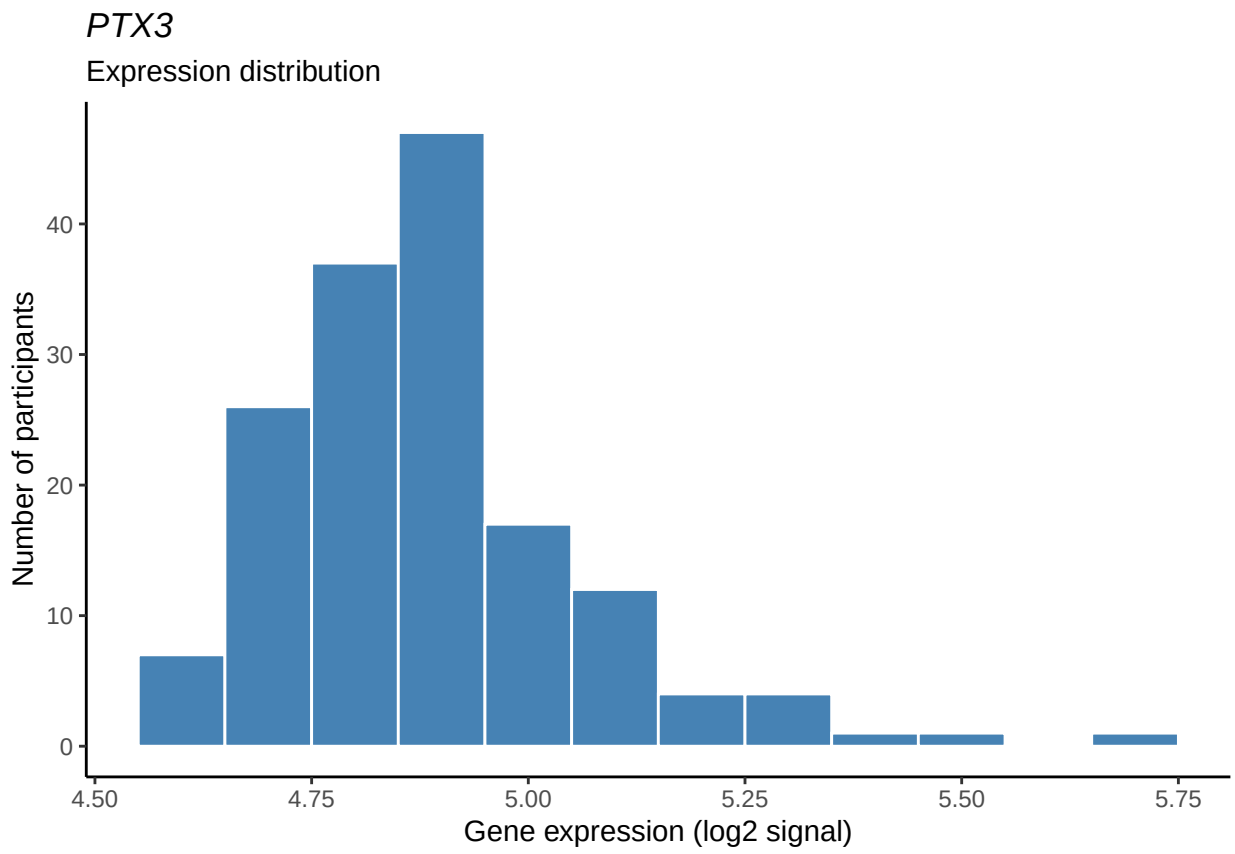

CXCR2
Expression by pregnancy outcome
Gene expression (log2 signal)
12.50
12.25
12.00
11.75
11.50
Normal
Preeclampsia
Pregnancy outcome
Vitamin D group
Below 30
30 or higher



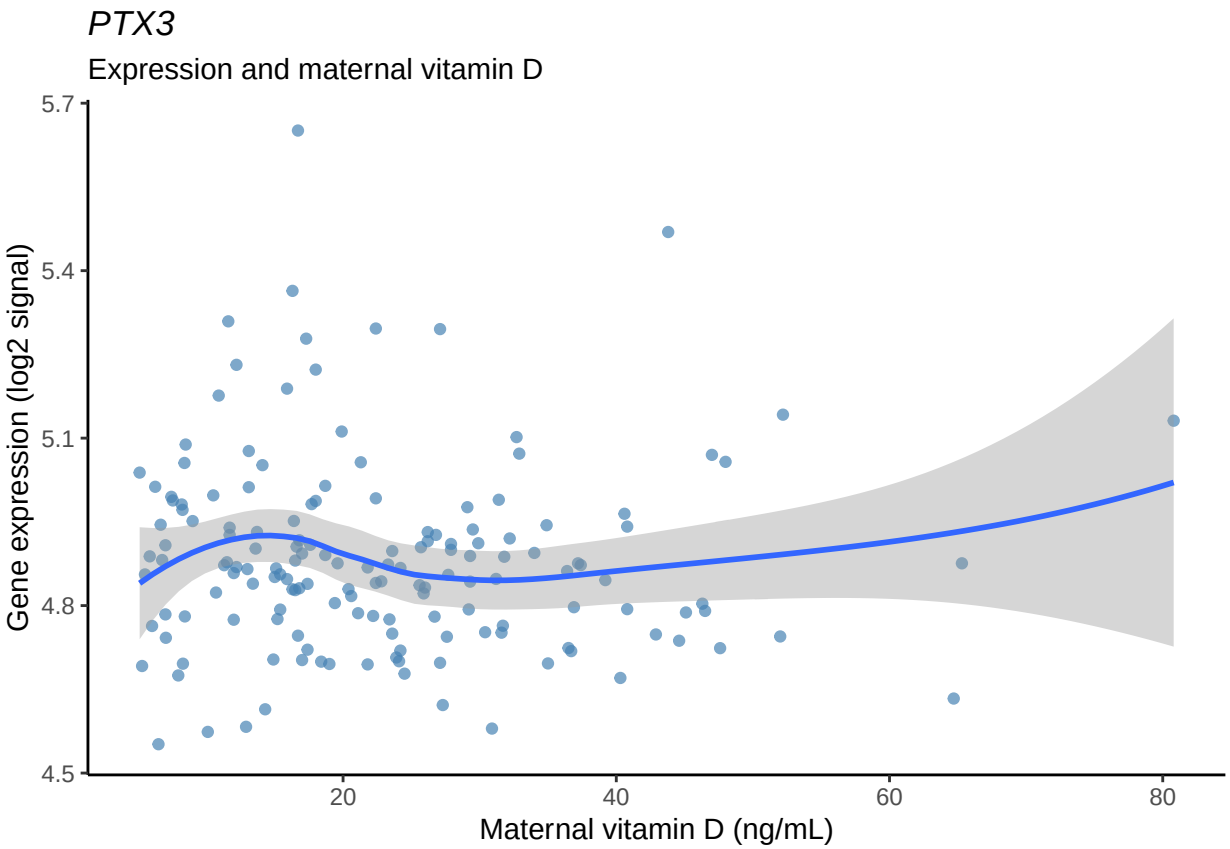

```
## `geom_smooth()` using formula = 'y ~ x'
```


CYP24A1
Expression and maternal vitamin D
Gene expression (log2 signal)
5.2
5.0
4.8
4.6
20
40
60
80
Maternal vitamin D (ng/mL)


CYP24A1
Expression by pregnancy outcome
Gene expression (log2 signal)
5.2
5.0
4.8
4.6
Normal
Preeclampsia
Pregnancy outcome
Vitamin D group
Below 30
30 or higher




```
## `geom_smooth()` using formula = 'y ~ x'
```


PTX3
Expression and maternal vitamin D
Gene expression (log2 signal)
5.7
5.4
5.1
4.8
4.5
20
40
60
80
Maternal vitamin D (ng/mL)

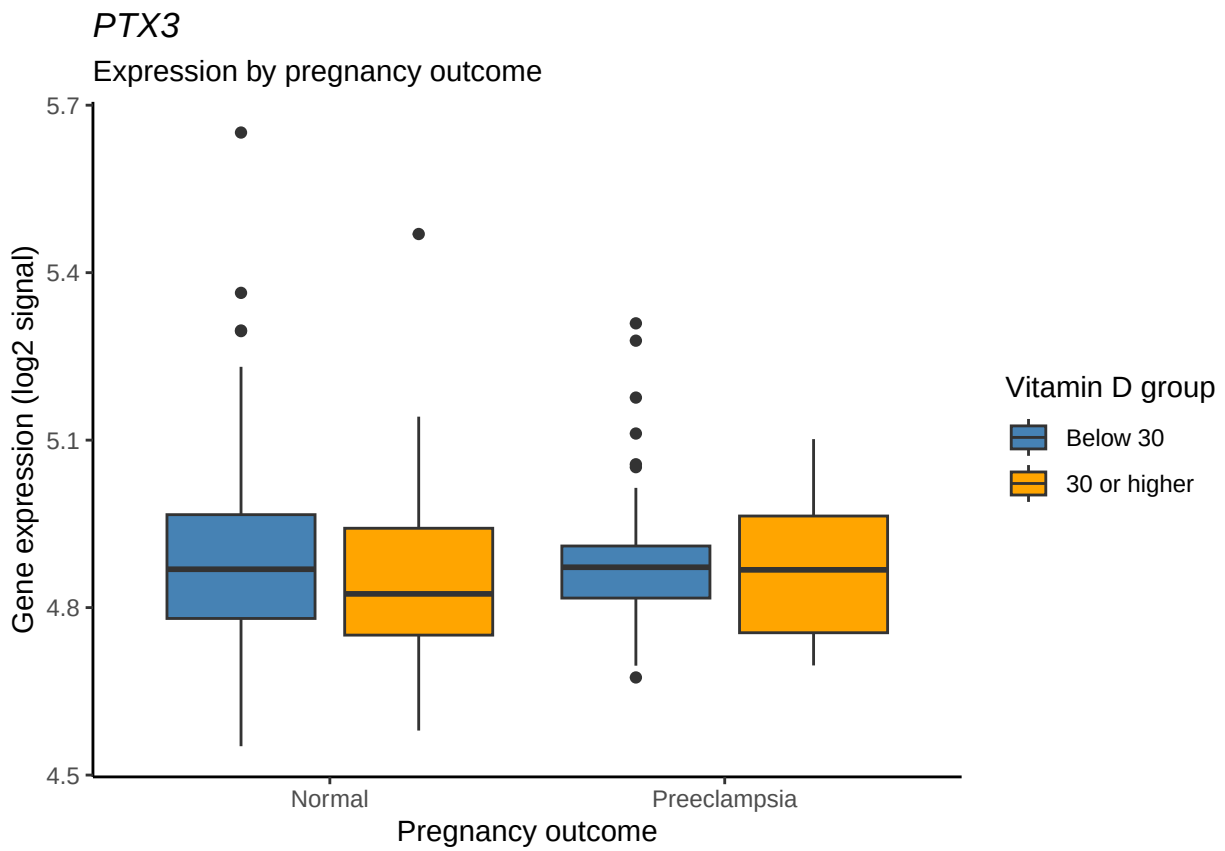


Table 1: Summary Statistics

```
# Separate participants by pregnancy outcome
normalData <- analysis[analysis$outcome == "Normal", ]
preeclampsiaData <- analysis[analysis$outcome == "Preeclampsia", ]

# Check the number of participants
nrow(normalData)
```

```
## [1] 110
```

```
nrow(preeclampsiaData)
```

```
## [1] 47
```

```
# Calculate mean and standard deviation of maternal age
mean(normalData$age_years)
```

```
## [1] 26.63182
```

```
sd(normalData$age_years)
```

```
## [1] 5.08356
```

```r
mean(preeclampsiaData$age_years)
```

```
## [1] 25.77687
```

```r
sd(preeclampsiaData$age_years)
```

```
## [1] 4.949298
```

```r
# Calculate mean and standard deviation of maternal BMI
mean(normalData$bmi)
```

```
## [1] 27.9924
```

```r
sd(normalData$bmi)
```

```
## [1] 7.662659
```

```r
mean(preeclampsiaData$bmi)
```

```
## [1] 31.15726
```

```r
sd(preeclampsiaData$bmi)
```

```
## [1] 8.028971
```

```r
# Calculate mean and standard deviation of maternal vitamin D
mean(normalData$vitamin_d)
```

```
## [1] 24.37455
```

```r
sd(normalData$vitamin_d)
```

```
## [1] 14.54883
```

```r
mean(preeclampsiaData$vitamin_d)
```

```
## [1] 19.72468
```

```r
sd(preeclampsiaData$vitamin_d)
```

```
## [1] 8.329801
```

Formatted Summary Statistics Table

```r
# Format continuous variables as mean (SD)
mean_sd <- function(x) {
  paste0(round(mean(x), 2), " (", round(sd(x), 2), ")")
}

# Format categorical variables as count (percentage)
count_pct <- function(data, group) {
  n <- sum(data$vitamin_d_group == group)
  paste0(n, " (", round(n / nrow(data) * 100, 1), "%)")
}

# Build Table 1
table1 <- data.frame(
  Variable = c("Maternal age (years)",
               "Maternal BMI (kg/m2)",
               "Vitamin D (ng/mL)",
               "Vitamin D below 30",
               "Vitamin D 30 or higher"),

  Normal = c(mean_sd(normalData$age_years),
             mean_sd(normalData$bmi),
             mean_sd(normalData$vitamin_d),
             count_pct(normalData, "Below 30"),
             count_pct(normalData, "30 or higher")),

  Preeclampsia = c(mean_sd(preeclampsiaData$age_years),
                   mean_sd(preeclampsiaData$bmi),
                   mean_sd(preeclampsiaData$vitamin_d),
                   count_pct(preeclampsiaData, "Below 30"),
                   count_pct(preeclampsiaData, "30 or higher"))
)

knitr::kable(table1,
             caption = "Maternal characteristics by pregnancy outcome")
```

Table 1: Maternal characteristics by pregnancy outcome

Variable	Normal	Preeclampsia
Maternal age (years)	26.63 (5.08)	25.78 (4.95)
Maternal BMI (kg/m2)	27.99 (7.66)	31.16 (8.03)
Vitamin D (ng/mL)	24.37 (14.55)	19.72 (8.33)
Vitamin D below 30	78 (70.9%)	41 (87.2%)
Vitamin D 30 or higher	32 (29.1%)	6 (12.8%)

10-Gene Heatmap

```r
# Load the heatmap package
library(pheatmap)

# Select 10 genes from the dataset
```

```r
heatmap_genes <- c(
  "CXCR2", "CYP24A1", "PTX3",
  "SOCS5", "SIRT3", "HSPA8",
  "CNR1", "TRIM24", "SLC1A5", "RBL2"
)

# Create an empty matrix for gene expression
heatmapData <- matrix(
  nrow = length(heatmap_genes),
  ncol = nrow(analysis)
)

# Assign gene names and sample IDs
rownames(heatmapData) <- heatmap_genes
colnames(heatmapData) <- analysis$geoID

# Add expression values for each gene
for (gene in heatmap_genes) {
  probe <- as.character(linker$PROBEID[linker$GENE == gene])
  heatmapData[gene, ] <- expression[analysis$geoID, probe]
}
```

Heatmap Tracking Bars

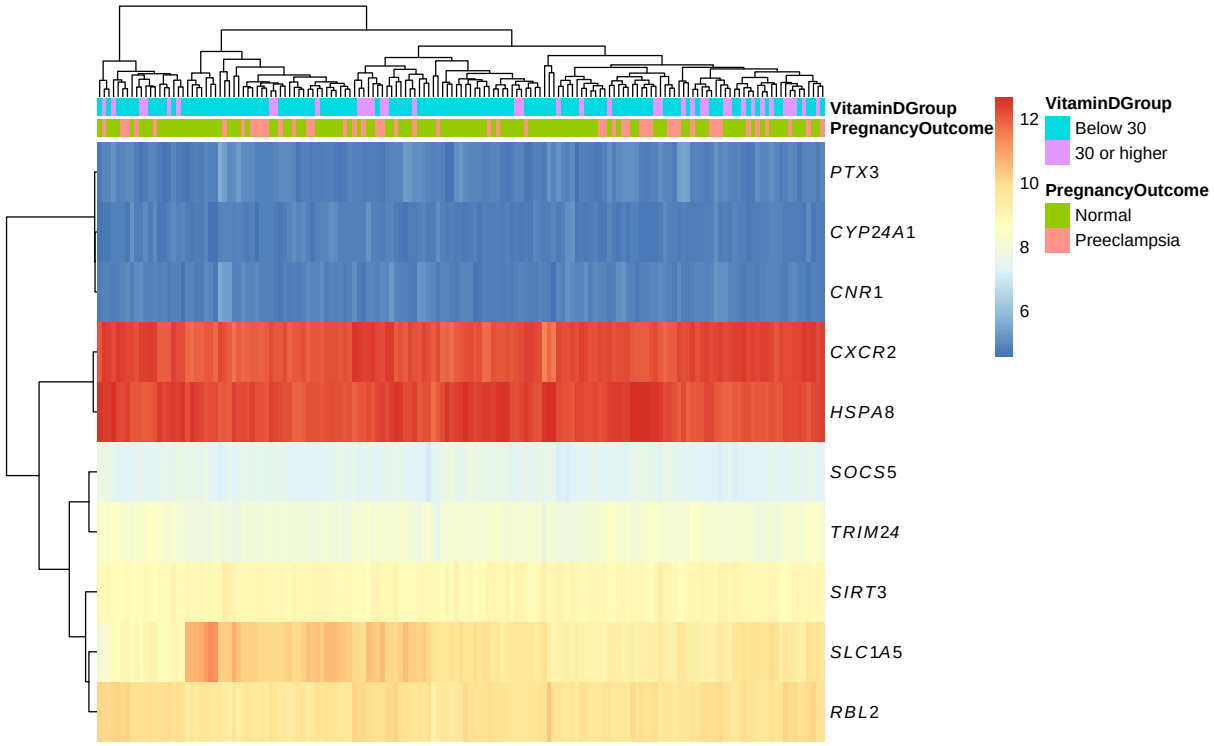
```r
# Create two categorical tracking bars
annotationData <- data.frame(
  row.names = analysis$geoID,
  PregnancyOutcome = analysis$outcome,
  VitaminDGroup = analysis$vitamin_d_group
)
```

Clustered 10-Gene Heatmap

```r
# Generate a heatmap with hierarchical clustering

pheatmap(
  heatmapData,
  clustering_distance_cols = "euclidean",
  clustering_distance_rows = "euclidean",
  annotation_col = annotationData,
  labels_row = expression(
    italic(CXCR2),
    italic(CYP24A1),
    italic(PTX3),
    italic(SOCS5),
    italic(SIRT3),
    italic(HSPA8),
    italic(CNR1),
    italic(TRIM24),
```

```
    italic(SLC1A5),
    italic(RBL2)
  ),
  show_colnames = FALSE
)
```



Additional Plot: Gene Expression Differences

```
# Identify participants in each pregnancy group
normal_ids <- analysis$geoID[analysis$outcome == "Normal"]
preeclampsia_ids <- analysis$geoID[analysis$outcome == "Preeclampsia"]

# Prepare a table for the mean differences
gene_diff <- data.frame(
  Gene = heatmap_genes,
  Difference = NA
)

# Calculate expression differences for each gene
for (gene in heatmap_genes) {
  gene_diff[gene_diff$Gene == gene, "Difference"] <-
    mean(heatmapData[gene, preeclampsia_ids]) -
    mean(heatmapData[gene, normal_ids])
}
```

Lollipop Plot

```
# Compare mean gene expression between pregnancy outcomes
ggplot(gene_diff, aes(x = Difference, y = Gene)) +
  geom_segment(aes(x = 0, xend = Difference,
                   y = Gene, yend = Gene),
               color = "steelblue", linewidth = 0.7) +
  geom_point(color = "steelblue", size = 3) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Gene expression differences by pregnancy outcome",
    x = "Mean difference: Preeclampsia - Normal (log2 signal)",
    y = "Gene"
  ) +
  theme_classic() +
  theme(axis.text.y = element_text(face = "italic"))
```

